Lan Pan

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RESEARCH INTERESTS

Computational cognitive science; human and machine decision-making; AI for science; behavioral data infrastructure; language-based measurement; LLM behavior and cognitive modeling.

EDUCATION

Southwestern University of Finance and Economics, China
Double Bachelor's Degree in Marketing and Finance, Sep 2015 - Jun 2019
GPA: 3.5, Rank: 6/105

Undergraduate theses:

- Marketing: The factors influencing purchase intention for new energy vehicles.
- Finance: An empirical study on the relationship between RMB exchange-rate fluctuations and the international balance of payments.

PUBLICATIONS

Pan, L.*, Xie, H.*, & Wilson, R. C. (2025). Large language models think too fast to explore effectively. *Advances in Neural Information Processing Systems, 38*, 102480-102508. https://doi.org/10.52202/085713-3428

Pan, L., & Musslick, S. (2026). Has someone already tested this? Construct-level hypothesis-literature matching. *Discovery Science 2026* (accepted).

Xie, H., Jagadish, A. K., **Pan, L.**, & Wilson, R. C. (2026). *Think-aloud reshapes automated cognitive model discovery beyond behavior* [Conference presentation]. 9th Annual Conference on Cognitive Computational Neuroscience (CCN). https://arxiv.org/abs/2605.05091

* Equal contribution.

MANUSCRIPTS AND WORKING PAPERS

"Behavioral-Computational Task Space of Cognitive Tasks." Manuscript in preparation.

RESEARCH EXPERIENCE

Research Assistant, NRD Lab, Georgia Institute of Technology
Supervisor: Robert C. Wilson; Hanbo Xie
Nov 2024 - Present

- Conduct research on large language models as computational models of human learning and decision-making.
- **Project 1 (NeurIPS 2025):** Led research evaluating whether large language models can explore effectively in an open-ended discovery task, benchmarked against large-scale human behavior. Identified systematic gaps in exploratory performance across model families and characterized their decision patterns using a compact strategy analysis.
- **Project 2 (ongoing):** Investigate cross-task generalization of LLMs trained on individual psychology tasks, analyzing similarities and differences across task-induced behaviors.

Research Collaborator
Collaborators: Akshay K. Jagadish and Younes Strittmatter, Princeton University
Jun 2026 - Present

- **Ongoing:** Develop an agent that answers researchers' queries by identifying relevant evidence across a cross-task collection of thousands of behavioral datasets and analyzing the selected datasets.

Research Assistant, Automated Scientific Discovery of Mind and Brain Lab, University of Osnabrueck
Supervisor: Sebastian Musslick
Dec 2025 - Present

- Work on automated scientific discovery systems for cognitive science.
- **Project 1 (Discovery Science 2026, accepted):** Developed a construct-level hypothesis-literature matching pipeline that retrieves papers, extracts scientific hypotheses, and assesses construct-level matches.
- **Project 2 (ongoing):** Develop a label-free event-state representation of cognitive-control paradigms to compare task structures across studies and test when differently named paradigms may rely on shared mechanisms.

PROFESSIONAL EXPERIENCE

Chengdu Jiaotou Huaizhou Xincheng Investment and Operation Co., Ltd., China
Investment and Operation
Nov 2021 - Jun 2023

Chengdu Youzhujialian Group Co., Ltd., China
Planning Management Trainee; Planning Manager
Jul 2019 - Nov 2021

ACADEMIC SERVICE

Reviewer, Frontiers of Computer Science, 2025

SKILLS

- Programming and data analysis: Python, pandas, NumPy, statistical modeling, machine learning workflows.
- Behavioral and cognitive modeling: regression, ANOVA, PCA, factor analysis, task-performance analysis, cross-task transfer analysis.
- LLM and NLP methods: LoRA fine-tuning, embeddings, semantic similarity, natural language inference, model deployment, quantization.
- Scientific infrastructure: reproducible analysis pipelines, dataset integrity checks, provenance tracking, agentic analysis workflows.
- Writing and tools: LaTeX, Git.

SELECTED COURSEWORK

Advanced Mathematics; Linear Algebra; Probability Theory; Statistics; Econometrics; Consumer Behavior; Market Research; Database Management Systems; Macroeconomics; Microeconomics; Corporate Finance; Investments; International Finance; Derivative Financial Instruments.